

Embedded-AI-Driven On-Site Traumatic Brain Injury Assessment Using Wireless Flexible Wearable Sensors for Real-Time Impact Force and Acceleration Estimation

Xuan Chen¹, Shuxun Wang¹, Yicheng Zhang, Yongping Ye, Jianbo Ye, Wei Li², *Member, IEEE*, Dibo Hou¹, and Yunqi Cao¹, *Member, IEEE*

Abstract—Traumatic brain injury (TBI) assessment is crucial for protecting the health of casualties involved in sudden head-impact incidents. However, traditional methods for assessing TBI are primarily based on bulky imaging equipment, which suffers from time lag, leading to misjudgment of the injury and missing the golden opportunity for treatment. Moreover, most existing Internet of Things (IoT)-aided sensor-based head impact detection studies monitor only a single biomechanical parameter, either the external head impact force or the head center-of-mass acceleration, thus offering an incomplete injury profile. This study proposed a method for rapidly acquiring on-site time-series data to support TBI assessment, which employed flexible piezoelectric sensors to detect head impact, whose electromechanical coupling was characterized by an equivalent-impedance model identified with the genetic algorithm (GA). The voltage signal from the sensor was transmitted to mobile devices via Bluetooth to estimate head-impact biomechanics. Specifically, the impact force on the wearer's head was converted from the voltage using the sequential quadratic programming (SQP) algorithm. Based on the impact force data, the acceleration of the center of mass of the head was predicted with a neural network containing the long short-term memory (LSTM) layer. The experiments were carried out on a head-neck dummy subjected to different levels of impact. The resulting models achieved coefficients of determination (R^2) of 0.935 for impact force (amplitude of 0–3 kN) and 0.848 for acceleration of the center of mass (amplitude of 0–120 g). The fabricated wireless flexible wearable sensor was installed on the dummy for performance validation, successfully demonstrating collision detection, impact force, and acceleration estimation for TBI assessment, which

can be conveniently deployed in helmets used in construction, sports, and emergency rescue, leveraging IoT technology for real-time dual-parameter head-impact estimation and medical-data synchronization to safeguard the golden treatment window and hopefully promote the efficient allocation of medical resources.

Index Terms—Acceleration prediction, flexible piezoelectric sensors, force optimization, head impact, Internet of Things (IoT), traumatic brain injury (TBI), wearable devices.

I. INTRODUCTION

TRAUMATIC brain injury (TBI) is an injury to brain tissue caused by blunt force or sharp force on the head, resulting in changes in brain structure or function, accompanied by temporary or permanent neurological dysfunction. TBI is one of the brain diseases with the highest rate of death and disability worldwide, caused by sudden collisions such as traffic accidents and violence, leading to direct trauma or irreversible effects on the patient's speech, behavior, and intellectual ability [1], [2]. Statistics show that there are more than 50 million cases of TBI worldwide each year, especially among children and adolescents, and males are about twice as likely as females to suffer a TBI [3], [4].

At present, the diagnosis of TBI mainly relies on large-scale medical equipment such as magnetic resonance imaging machines and head computerized tomography, which cannot provide timely and accurate on-site medical diagnostic data, making it difficult for patients to judge their injuries in time. It may even make wrong judgments and miss the golden time for treatment [5]. Given the necessity of real-time assessment of TBI, Internet of Things (IoT)-aided wearable devices can provide a platform support for the rapid detection of TBI, guaranteeing the quality of medical emergencies [6], [7].

TBI can be categorized into diffuse brain injury and focal brain injury. Diffuse brain injury is usually caused by inertial forces such as linear acceleration and rotational angular acceleration acting on the head-neck system [8], [9]. Focal brain injury is usually caused by an impact force acting directly on the head. To measure acceleration of head during collisions accurately, IoT-aided methods directly based on inertial sensors were proposed by Liu et al. [10], Kuo et al. [11], Cecchi et al. [12], and Patton et al. group [13], but

Received 11 August 2025; revised 28 September 2025 and 24 December 2025; accepted 12 January 2026. Date of publication 19 January 2026; date of current version 9 March 2026. This work was supported in part by Zhejiang Provincial Natural Science Foundation of China under Grant TGY23F030004, in part by Zhejiang Provincial Department of Emergency Management under Grant 2025YJ035, and in part by the National Natural Science Foundation of China under Grant 62273304. (Corresponding author: Yunqi Cao.)

Xuan Chen, Shuxun Wang, Yicheng Zhang, Dibo Hou, and Yunqi Cao are with the State Key Laboratory of Industrial Control Technology, College of Control Science and Engineering, Zhejiang University, Hangzhou, Zhejiang 310027, China (e-mail: caoyunqi@zju.edu.cn).

Yongping Ye and Jianbo Ye are with Zhejiang Emergency Management Digital and Technology Center, Hangzhou, Zhejiang 310007, China (e-mail: 461474897@qq.com; leavebo@163.com).

Wei Li is with the College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing, Jiangsu 210046, China (e-mail: wei.li@njupt.edu.cn).

Digital Object Identifier 10.1109/JIOT.2026.3655010

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the detection methods suffered from wear comfort issues and significant detection errors, which also could not measure the impact force. To measure the impact force during collisions accurately, IoT-aided methods based on flexible piezoelectric sensors were proposed by Sahu et al. [14], Merrell et al. [15], and Li et al. group [16], which achieved good wear comfort, but the measurement accuracy still needed to be improved and the methods could not measure the head acceleration at the same time. Overall, existing studies only considered a single physical variable of impact force or acceleration during collisions, and they could not quickly and comprehensively assess TBI.

In this article, a TBI preliminary assessment method using a wireless flexible wearable sensor to estimate the impact force and acceleration during head collisions was proposed. The proposed IoT system framework begins with a helmet-mounted polyvinylidene fluoride (PVDF) sensor that converts a head impact into a high-frequency voltage waveform. A tiny MCU digitizes the signal and streams the raw samples over a Bluetooth link to a nearby mobile device. The mobile device buffers the packets, runs an on-device model that transforms voltage into impact force and head-center acceleration, and assesses TBI preliminarily with audible or haptic alerts. The relevant results will be uploaded to the cloud with the help of the Internet, and the data will be synchronized with the medical institutions. In the experiment, based on the scenario where the head was subjected to an impact, a drop tower equipped with a dummy head was designed to simulate head impacts of different intensities to determine the applicability of the PVDF sensors in evaluating different impact levels. Then, the output voltage of the PVDF sensor, the impact force on the dummy head measured from the impact sensor, and the center-of-mass acceleration of the dummy head measured from the acceleration sensor during the collision process were collected. The sensor electromechanical coupling model was fit using the genetic algorithm (GA) based on the voltage and impact force data, upon which the impact force was subsequently estimated using the sequential quadratic programming (SQP) algorithm. To complete the prediction of acceleration, a neural network containing the long short-term memory (LSTM) layer was trained based on the impact force and acceleration data. Finally, the performance of the independently designed wireless flexible wearable sensor was validated on the dummy head. This work contributes to presenting the first scheme for the real-time dual-parameter head-impact estimation and designing the wireless flexible wearable sensor, which provides an innovative and feasible solution for comprehensive on-site assessment of TBI, safeguards the golden treatment window, and promotes the efficient allocation of medical resources.

II. EXPERIMENTAL SECTION

A. Fabrication of PVDF Sensor

This study used PVDF piezoelectric electret film as the impact force detection-sensitive element. The configuration design of the PVDF sensor is shown in Fig. 1(a). First, a PVDF film (dimension of 30×30 mm, thickness of

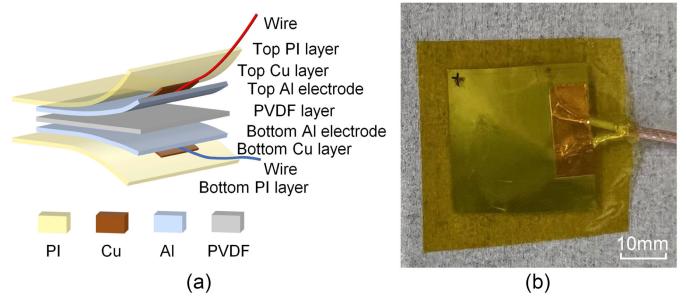


Fig. 1. Design and fabrication of PVDF sensor. (a) Layered structural design. (b) Physical sensor.

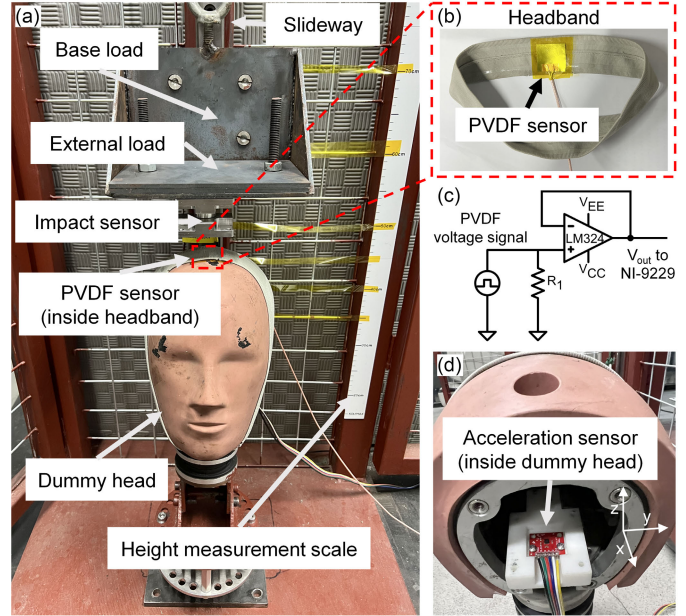


Fig. 2. Experimental setup. (a) Collision device design and sensor deployment. (b) Headband with PVDF sensor. (c) Voltage following circuit. (d) Deployment position and orientation of the acceleration sensor.

50 μm) with Aluminum films as electrodes covering both sides was fabricated by KEXIN ELECTRONIC. Then, the wires were attached to the Aluminum electrode using two conductive copper tapes (dimension of 7.5×15 mm) to ensure good contact. After that, the PVDF film containing the electrodes, copper tapes, and wires was encapsulated with two polyimide (PI) films to provide the PVDF film with mechanical robustness. The experimentally fabricated PVDF sensor, which can sense impact and output voltage signals, is shown in Fig. 1(b).

B. Experimental Setup

The head impact experiment was carried out on the experimental setup shown in Fig. 2(a), which consists of four parts: 1) the slideway; 2) the dummy head; 3) the weight (containing base load and external loads); and 4) the signal generation and acquisition module. To simulate different head impact levels, the experiment used base load without external load to fall from heights of 20, 40, 60, 80, 100, and 120 cm, as well as base load with 0, 2, 4, 6, 8, and 10 units of

external load to fall from a height of 20 cm, to collide with the Hybrid III 50th Male dummy head model. Each condition was tested in five independent trials. The dummy head model closely resembles the human head in size, shape, and mass. Originally developed for automotive crash testing, the Hybrid III 50th Male dummy reflects human collision dynamics well. Its compliant outer integument reproduces the bulk stiffness and frictional properties of human skin and soft tissue, while the internal construction incorporates a rigid skull surrogate with calibrated moments of inertia to mimic cranial mass distribution [17], [18]. When coupled with the Hybrid III neck, the head–neck system reproduces human-like global kinematics in impact testing. The mass of the base load (including the impact sensor and the slider) was 5.234 kg, the average mass of the external loads was 1.021 kg, and the weight (containing base load and external loads) was connected to the slideway via the slider to maintain collision consistency. Based on this device, the output voltage of the PVDF sensor (amplitude of 0–12 V), the impact force on the head (amplitude of 0–3 kN), and the acceleration of the center of mass of the head (amplitude of 0–120 g) were collected during the collision process.

To characterize the PVDF sensor, the voltage from the sensor and the impact force applied to the sensor must be measured simultaneously. Therefore, the PVDF sensor was tightly attached to the top of the dummy head with the help of the headband as shown in Fig. 2(b). It was hit by the frontal impact of the falling weight, containing base load and external loads, and generated the corresponding voltage signals. The voltage signal can be measured using the voltage following circuit shown in Fig. 2(c) where $V_{CC} = +12$ V, $V_{EE} = -12$ V, and $R_1 = 20$ k Ω . The voltage following circuit was constructed using an operational amplifier (LM324, Texas Instruments, Inc.) and a bias resistor of 20 k Ω to provide a dc bias and to control the amplitude of the voltage signals within 12 V. To measure impact force signals, an impact sensor (6012, Lindmc, Inc.) was fixed at the bottom of the falling weight to directly detect the collision impact force, which was coupled with the corresponding charge amplifier (6901, Lindmc, Inc.) to realize the voltage signal generation, and the output signal of the impact sensor were 0–10 V, which corresponded to impact force of 0–10 kN.

A neural network was trained to predict acceleration using impact force, and acceleration data of the dummy head must be measured simultaneously during collisions to train the neural network model. Therefore, a triaxial acceleration sensor (ADXL377, Analog Devices, Inc.) was deployed inside the dummy head with the help of a 3D-printed base as shown in Fig. 2(d), which was rigidly connected to the dummy head to measure the center-of-mass acceleration accurately, having a significant importance in TBI assessment.

The signals generated by the PVDF sensor, the impact sensor, and the acceleration sensor were acquired through an analog input module (NI-9229, National Instruments Corporation) and sent to a personal computer for data processing with the help of a real-time controller (cRIO-9039, National Instruments Corporation).

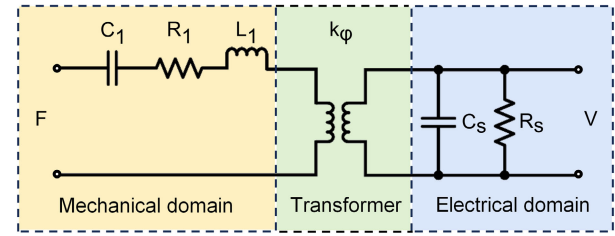


Fig. 3. Electromechanical coupling model of PVDF sensor.

C. Sensor Modeling and Impact Force Optimization

The PVDF sensor has good dynamic response characteristics and can convert the instantaneous force applied to it into an electrical output. Due to its viscoelasticity in the mechanical domain and similar capacitive power generation characteristics in the electrical domain, an electromechanical coupling model as shown in Fig. 3 is used to describe the PVDF sensor system by transforming the spring-mass-damper model into an equivalent circuit in the mechanical domain and characterizing it with the RLC model. In the mechanical domain, the model of PVDF film can be described as a spring-mass-damper. The spring is represented by the capacitance C_1 , the mass is represented by the inductance L_1 , which means the mechanical inertia of the film, and the damper is defined by the resistance R_1 , which means the mechanical damping force generated by the viscosity of the film material and air. In the electrical domain, the PVDF film can be modeled as a parallel-plate capacitor with capacitance C_s and resistance R_s . The capacitance C_s is defined by the thickness of the PVDF film, the electrode area, and the dielectric constant of the PVDF. The resistance R_s is determined by the parallel connection of the film's internal resistance and the measurement circuit's external resistance. The transformer represents the conversion between the mechanical domain and the electrical domain. It is defined by the electromechanical coupling constant k_ϕ , which measures the efficiency of the conversion between the electrical energy and the mechanical energy. Finally, the transfer function of the sensor system between the voltage V generated by the PVDF sensor and the impact force F measured by the impact sensor can be obtained based on the equivalent circuit as follows:

$$G(s) = \frac{V(s)}{F(s)} = \frac{s}{a_3 s^3 + a_2 s^2 + a_1 s + a_0} \quad (1)$$

where the parameters in the transfer function are defined as follows: $a_3 = L_1 C_s / k_\phi$, $a_2 = L_1 / R_s k_\phi + R_1 C_s / k_\phi$, $a_1 = R_1 / R_s k_\phi + C_s / C_1 k_\phi + k_\phi$, and $a_0 = 1 / C_1 R_s k_\phi$.

GA with strong global search ability was adopted to optimize the four parameters in the transfer function of the sensor system to avoid falling into the local optimal solution. Specifically, to address the immense magnitude disparity among the parameters, scaling factors were introduced to normalize the search space, ensuring the algorithm treats all variables with equal weight. A sufficiently large population size was employed to maximize exploration density across the multidimensional solution space, prioritizing the breadth of search. Furthermore, strict convergence criteria were

TABLE I
CONFIGURATION PARAMETERS OF THE PROPOSED LSTM NETWORK

Category	Parameter	Description
Network Architecture	Input	Sequence (1-dim)
	Hidden Layers	$2 \times \text{LSTM (100 units)}$
	Dense Layers	$100 \text{ units} \xrightarrow{\text{ReLU}} 1 \text{ units}$
	Output	Sequence Regression
	Loss Function	Acceleration MSE
Hyperparameters	Optimizer	Adam
	Initial Learning Rate	0.01
	Mini-Batch Size	10
	Max Epochs	500
	Gradient Threshold	1
Training Process	Data Split Strategy	Train : Val : Test = 10 : 1 : 3
	Validation Patience	5 epochs

The network utilizes two LSTM layers followed by dense layers to map impact force to acceleration. Data augmentation includes random amplitude scaling, circular shifting, and Gaussian noise injection.

established, characterized by a low function tolerance and a high stall generation limit, compelling the algorithm to persist in fine-tuning until the residual error was minimized to a negligible level.

After obtaining the sensor system model, the experiment verified the inverse process of solving the impact force applied to the sensor from the sensor output voltage with the help of SQP, which can convert the nonlinear optimization problem into a quadratic programming problem and solve the impact force signal with high efficiency. To ensure robust convergence and high-precision reconstruction, the optimization algorithm was configured with sufficiently large limits for both iterations and function evaluations, thereby avoiding premature termination. Furthermore, stringent criteria were imposed on optimality and step tolerances to ensure the solver persisted until the solution reached a high degree of accuracy.

D. Head Center-of-Mass Acceleration Prediction

To predict the center-of-mass acceleration of the head based on the impact force on the head, a kinetic model of the head needs to be obtained. Modeling the head model directly requires a large number of model parameters. Based on the strong correlation between the center-of-mass acceleration of the head and the impact force on the head [19], [20], a neural network model with LSTM layers was used to predict the center of mass acceleration of the head using the impact force on the head, which is characterized by long-term dependency capture, suitable for processing complex time series and forecasting time series. The network architecture and hyperparameters were determined through an iterative tuning process aimed at balancing model complexity and generalization capability. Key parameters, such as the number of LSTM layers and hidden units, were optimized to capture temporal dependencies effectively. To ensure training stability and mitigate overfitting, strategies including gradient thresholding, data augmentation, and early stopping based on validation loss were incorporated. The final optimized configuration is detailed in Table I.

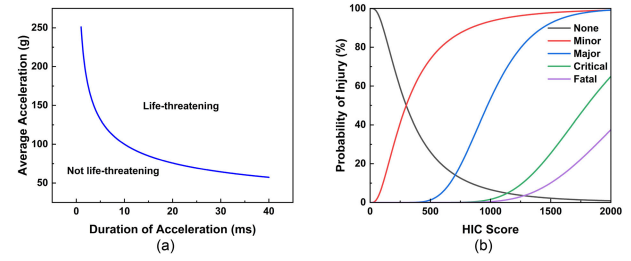


Fig. 4. Medical injury criteria from acceleration outputs. (a) Schematic representation of the WSTC showing the relationship between average acceleration and time duration for head injury thresholds. (b) Schematic representation of the expanded Prasad–Mertz curves illustrating the probability of head injury severity levels based on HIC values.

E. Selection of Medical Assessment Indicators

In this study, the peak acceleration (a_{peak}) [21], [22], [23], [24], [25], [26] and head injury criterion (HIC) [21], [27], [28], [29], [30] metrics for quantitative assessment of TBI were adopted to complete the injury assessment process as follows:

$$a_{\text{peak}} = \max \|\vec{d}(t)\| \quad (2)$$

$$\text{HIC} = \max_{t_1, t_2} \left\{ \left[\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \|\vec{d}(t)\| dt \right]^{2.5} (t_2 - t_1) \right\}. \quad (3)$$

For the calculation of the HIC metric, the time interval $t_2 - t_1$ is fixed at 15 ms, and the value is determined automatically using a sliding window algorithm.

F. Medical Evaluation Criteria of TBI Severity

For focal brain injury assessment, impact force data are essential for observing force distribution on the head and serve as a critical basis for assessing head injury. While peak impact force is also a valuable metric, the criteria for evaluating it vary significantly by application scenario.

For diffuse brain injury assessment of the medical assessment indicators concerning the acceleration outputs mentioned above, a_{peak} can be calculated to quickly assess for TBI based on the Wayne state tolerance curve (WSTC) [31], as shown in Fig. 4(a). The HIC is used to estimate the probability of different injury levels based on the expanded Prasad–Mertz curves [32], as shown in Fig. 4(b), which correlates well with the likelihood of brain injury.

The output voltage of the PVDF sensor was converted into the impact force applied on the head through the sensor's electromechanical coupling model. Then, the acceleration of the head's center of mass was predicted using a neural network that models the head's dynamics, enabling the determination of the impact force and acceleration during the collision. Based on the above indicators and relevant criteria, a preliminary diagnosis of TBI can be made.

G. IoT Embedded System Design and Construction

The IoT embedded system was designed and fabricated as shown in Fig. 5(a), which contains a signal transmitter module deployed on the head and a signal receiver module like smartphones or personal computers.

The signal transmitter module consists of a 5 V lithium battery, a 5 to ± 12 V dc converter board, a signal processing circuit, an STM32F103C8T6 microcontroller, a VG6328A Bluetooth, and a PVDF sensor, which realizes the signal generation, acquisition, and transmission functions. The signal receiver module realizes voltage signal acquisition and threshold detection, voltage signal slicing and smoothing, impact force signal optimization, acceleration signal prediction, and diagnosis of TBI by calculating medical indicators.

The signal processing circuit in the signal transmitter module mainly realizes the functions such as following, offsetting, scaling, and inverting of the voltage to be collected, and the specific circuit design [19] is shown in Fig. 5(b), which utilizes the LM324 to realize the voltage matching. The values of the parameters in Fig. 5(b) are shown below: $V_{CC} = +12$ V, $V_{EE} = -12$ V, $R_1 = R_4 = 10$ k Ω , $R_2 = R_3 = R_5 = R_6 = 100$ k Ω .

In practice, the PVDF sensor can detect the impact and generate the corresponding voltage signal. Then, the voltage signal can be transmitted to a mobile device through Bluetooth, where the impact force and acceleration estimation, as well as the corresponding damage indicator calculation, evaluation, and alarm, are carried out. The data provided by the wearable system will be uploaded to the cloud with the help of the Internet and synchronized with the medical institution, helping allocate healthcare resources and improve the hospital's triage efficiency.

III. RESULTS AND DISCUSSION

A. Signal Acquisition and Preprocessing

To explore the relationship between the voltage output of the PVDF sensor, the impact force on the head, and the acceleration of the center of mass of the head in a single collision, and to reduce the influence of extraneous signals on the results, it is necessary to slice and Gaussian-smooth the data collected in the experiment. The comparisons of preprocessed waveforms are shown in Fig. 6. In Fig. 6(a) and (b), the PVDF sensor outputs high-frequency oscillating voltage signals of decreasing amplitude, and the impact sensor measures bimodal force signals due to the relative motion between the weight and the head-neck system during collisions, which are consistent with the results found in [33]. Disregarding the effect of damping, the weight can be described by mass m_1 , and the head-neck system can be described by mass m_2 and spring k . In Fig. 6(b), as the mass of the falling weight m_1 increases, the duration of the signals from the PVDF sensor and the impact sensor also increases. Referring to the following equation:

$$\begin{cases} \omega \approx \sqrt{\frac{k}{m_1 + m_2}} \\ T = \frac{2\pi}{\omega} \end{cases} \quad (4)$$

where ω is the inherent frequency of the collision mode and T is the vibration period of the mode. Therefore, an increase in the mass of the falling weight m_1 leads to a decrease in ω , resulting in an increase in T , namely, an increase in the signal duration. Additionally, as the falling height or the mass

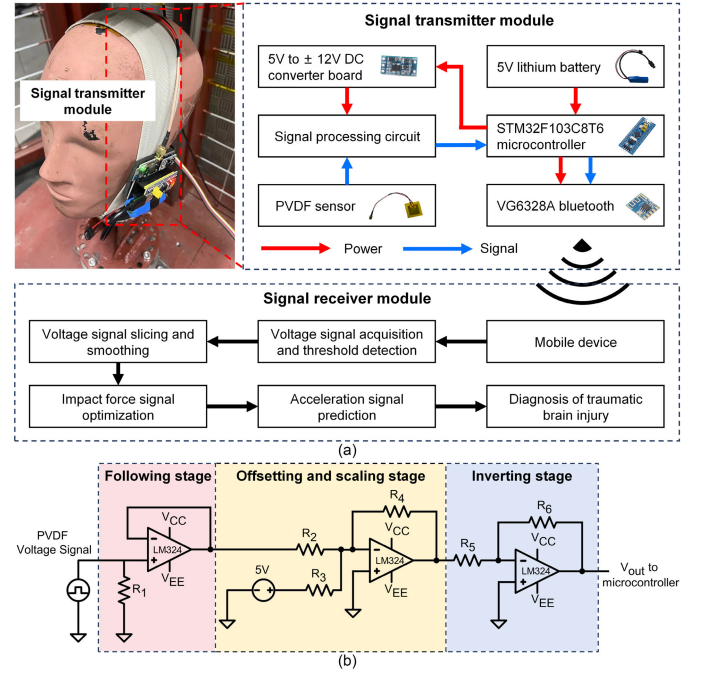


Fig. 5. Design of the IoT embedded system. (a) Laboratory deployment of the IoT embedded system and the IoT system framework incorporating a signal transmitter module and a signal receiver module connected over Bluetooth. (b) Voltage signal processing circuit of the PVDF sensor with following, offsetting, scaling, and inverting stages for voltage matching.

of the weight increases, namely, the level of the collision increases, the amplitude and the rate of change of the impact force signals will also increase accordingly, leading to an increase in the amplitude of the corresponding voltage signals. It can be seen that the PVDF sensor can detect and distinguish different levels of collision and output the corresponding voltage waveforms.

The deployment position and orientation of the acceleration sensor are shown in Fig. 2(d). From Fig. 6(c) and (d), the z -axis is the main direction of the acceleration component of the dummy head during collisions, and the acceleration amplitude in this direction is larger. In contrast, the acceleration components in the x -axis and y -axis directions are slight, which may be caused by head-neck coupled rotation and the vibration of the dummy head. In Fig. 6(e) and (f), the triaxial acceleration is converted to the center-of-mass acceleration by calculating the Euclidean L2 norm to calculate the medical indicators as in (2) and (3). A head impact with a bimodal waveform produces a corresponding bimodal acceleration, the amplitude of which increases as the impact force increases, and a slight acceleration can be briefly maintained after the end of the collision [11].

B. Model Parameter Fitting and Impact Force Optimization

To solve the parameters of the sensor model, the sum of squares of the residuals between the actual output voltage of the sensor V_{actual} and calculated voltage by the optimized sensor model $V_{calcd.}$ was used as the following fitness function:

$$f(a_0, a_1, a_2, a_3) = \sum (V_{actual} - V_{calcd.}(a_0, a_1, a_2, a_3))^2 \quad (5)$$

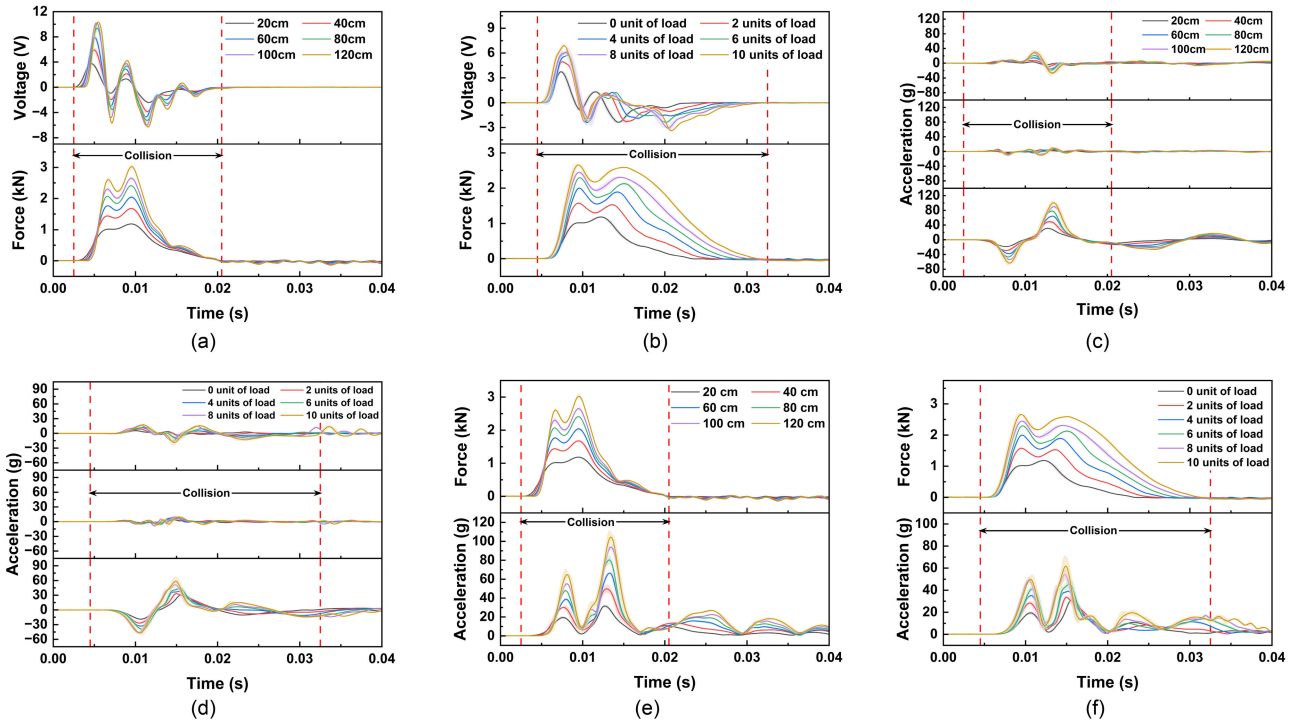


Fig. 6. Waveform comparisons of (a) and (b) voltage from PVDF sensor and force from impact sensor, (c) and (d) triaxial acceleration (from top to bottom: x-axis, y-axis, and z-axis data) from acceleration sensor, and (e) and (f) force from PVDF sensor and acceleration (the Euclidean L2 norm of the x-, y-, and z-axis acceleration data) measured by NI-9229 at heights of 20, 40, 60, 80, 100, and 120 cm using base load without external load or at the height of 20 cm using base load with 0, 2, 4, 6, 8, and 10 units of external load, to simulate different head impact levels. The portions between the red dashed lines are the corresponding waveforms generated when collisions occurred on the dummy head.

where V_{actual} is the actual voltage of the PVDF sensor measured by NI-9229 and $V_{\text{calcd.}}$ is calculated from the actual impact force based on the sensor model containing the fit parameters a_0 , a_1 , a_2 , and a_3 . The fitness function reflects the goodness of fitting the system parameters based on Fig. 3.

The parameters of the model were optimized using GA. Waveform comparisons of the actual voltage of the PVDF sensor and the voltage calculated by the optimized model are shown in Fig. 7(a)–(c). It can be seen that under the conditions of different falling heights and masses of the weights, the calculated and actual voltages are in good agreement, with an average MSE of 0.359 and an average R^2 of 0.879. The MAE and RMSE of the voltage are presented in Fig. 7(d). From the voltage results, it can be seen that the MAE of the voltage is controlled within 0.650 V and the RMSE of the voltage is controlled within 0.950 V. Across the different experimental conditions, the MAE expressed as a percentage (P-MAE) of the peak voltage ranges from 3.772% to 6.105%. The parameters in the transfer function were finally solved as follows: $a_3 = 6.469 \times 10^{-14}$, $a_2 = 1.620 \times 10^{-7}$, $a_1 = 3.916 \times 10^{-4}$, and $a_0 = 1.609 \times 10^2$.

Using the sensor model obtained above, inverse optimization was carried out based on the actual output voltage of the PVDF sensor using SQP to estimate the impact force. The sum of squares of the residuals between the actual output voltage of the PVDF sensor V_{actual} and the calculated voltage by the optimized impact force $V_{\text{calcd.}}$ was used as the following fitness

function:

$$f(F) = \sum (V_{\text{actual}} - V_{\text{calcd.}}(F))^2 \quad (6)$$

where V_{actual} is the actual voltage of the PVDF sensor measured by NI-9229 and $V_{\text{calcd.}}$ is calculated from the optimized impact force based on the sensor model obtained above. The fitness function reflects the goodness of optimizing the impact force on the head.

Waveform comparisons of the actual impact force and the optimized impact force are shown in Fig. 7(e)–(g). It can be seen that under the conditions of different falling heights and masses of the weights, the optimized and actual impact forces are in good agreement, with an average MSE of 0.040 and an average R^2 of 0.935. In Fig. 7(h), the MAE of the optimized impact force is controlled within 0.450 kN, and the RMSE of the optimized impact force is controlled within 0.500 kN. Across the different experimental conditions, the P-MAE of the peak force ranges from 2.682% to 13.969%. As the height or weight increases, the MAE and RMSE of the voltage and force tend to increase gradually in most cases, which may be due to the larger amplitude of the voltage or the impact force itself.

To visually assess the goodness of the impact force optimization, the force peak was calculated as a function of the drop height and weight via linear fitting to obtain the fitting results shown in Fig. 7(i) and (j), and Table II. The similarity between the actual and optimized data is analyzed in Fig. 7(k) and (l), and Pearson coefficients (r) are 0.954 and 0.998,

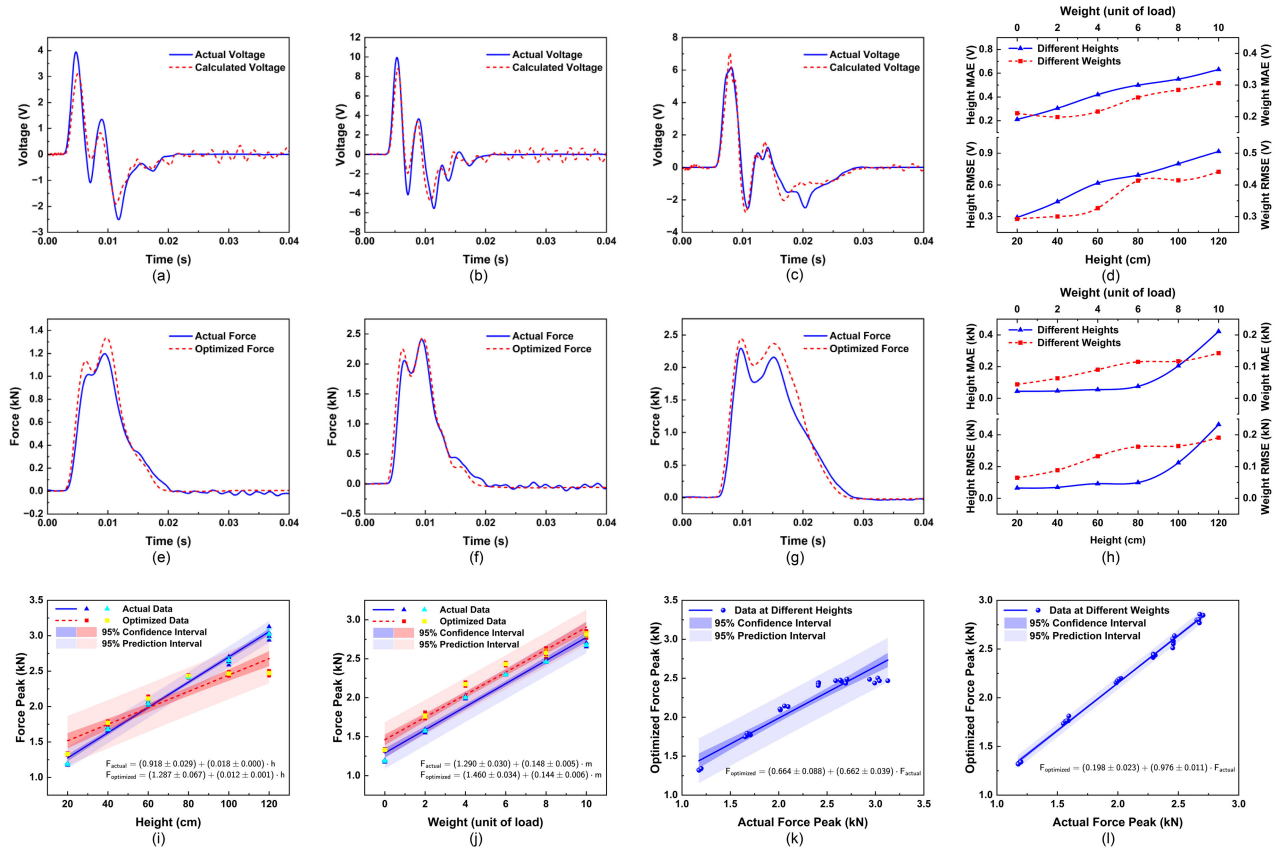


Fig. 7. Results of model parameter and impact force optimization. Voltage was calculated by the optimized model (a) at the height of 20 cm without external load, (b) at the height of 80 cm without external load, and (c) at the height of 20 cm with six units of external load, compared with the actual voltage. (d) MAE and RMSE of voltage were calculated at different heights and weights to assess how the model parameter optimization performance varied with drop height and weight. Impact force was optimized (e) at the height of 20 cm without external load, (f) at the height of 80 cm without external load, and (g) at the height of 20 cm with six units of external load, compared with the actual impact force. (h) MAE and RMSE of the impact force were calculated at different heights and weights to assess how the impact force optimization performance varied with drop height and weight. (i) and (j) Linear curves for the force peak with 95% confidence intervals and 95% prediction intervals were fit at different heights or weights, overlaid on scatter points showing means $\pm$ standard deviation. (k) and (l) Pearson correlation analysis was visualized using calibration plots incorporating 95% confidence intervals and 95% prediction intervals at different heights or weights, with correlations computed between the optimized and actual force peak.

TABLE II
RESULTS OF LINEAR CURVE FITTING FOR ACTUAL
AND OPTIMIZED IMPACT FORCE

Independent Variable	Metric	Pearson r	R ²
Height	Actual Peak Force	0.994	0.988
	Optimized Peak Force	0.931	0.866
Weight	Actual Peak Force	0.984	0.969
	Optimized Peak Force	0.980	0.960

Pearson coefficients (r) quantify the linear association between impact force peak and independent variables of the scatter points, while R² denotes the coefficient of determination of the linear fit.

respectively, which verifies the feasibility of estimating the impact force on the head from the output voltage of the PVDF sensor by the electromechanical coupling model. Based on the waveform comparisons and the quantitative analysis of the results, it can be preliminarily inferred that the impact force optimization results remain acceptable.

C. Acceleration Prediction and Assessment of Medical Indicators

A neural network model containing LSTM layers was trained with impact force as the input data and head

center-of-mass line acceleration as the output data. In the test set, the mean value of MSE is 13.154 and the mean value of R² is 0.916. Across the different experimental conditions, the P-MAE of the peak acceleration ranges from 2.455% to 5.646% in the test set.

The impact force data estimated using the output voltage of the PVDF sensor was used as an input to the trained model, and the corresponding acceleration data can be predicted, completing the signal conversion process from voltage to impact force to acceleration. Waveform comparisons of the actual acceleration and the acceleration predicted by the neural network are shown in Fig. 8(a)–(c). It can be seen that under the conditions of different falling heights and masses of the weights, the predicted and actual acceleration are in good agreement, with an average MSE of 28.213 and an average R² of 0.848. Notably, this R² value surpasses those reported in other studies, such as SWA-C (R² = 0.510) [10] and SIM-G (R² = 0.685) [13]. In Fig. 8(d), the MAE of the predicted acceleration is controlled within 6.000 g, and the RMSE of the predicted acceleration is controlled within 9.500 g. Across the different experimental conditions, the P-MAE of the peak acceleration ranges from 3.820% to 9.656%.

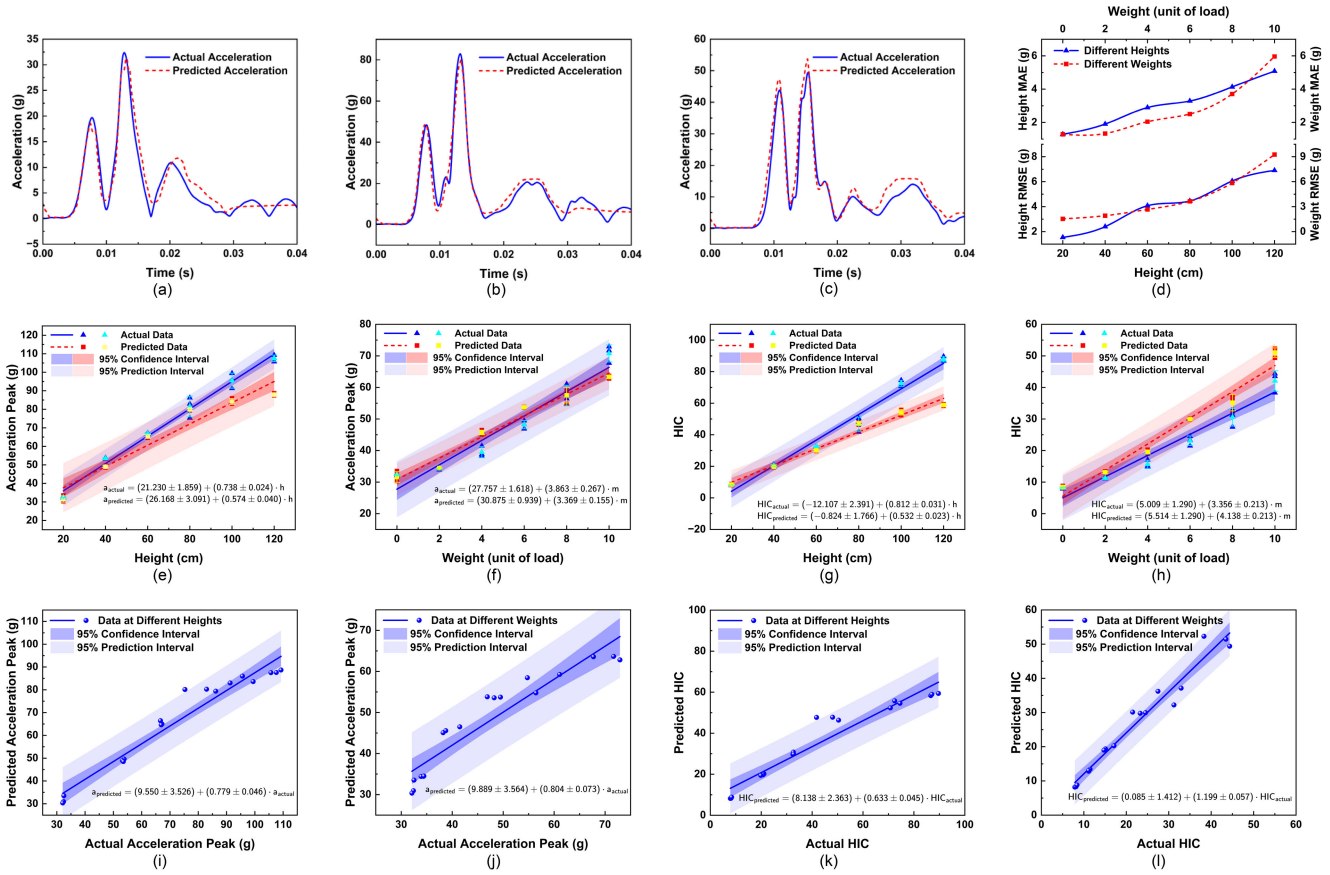


Fig. 8. Results of acceleration prediction. Acceleration was predicted by the neural network model (a) at the height of 20 cm without external load, (b) at the height of 80 cm without external load, and (c) at the height of 20 cm with six units of external load, compared with the actual acceleration. (d) MAE and RMSE of acceleration were calculated at different drop heights and weights to assess how the acceleration prediction performance varied with the drop height and weight. (e) and (f) Linear curves for the acceleration peak with 95% confidence intervals and 95% prediction intervals were fit at different heights or weights, overlaid on scatter points showing means $\pm$ standard deviation. (g) and (h) Linear curves for HIC with 95% confidence intervals and 95% prediction intervals were fit at different heights or weights, overlaid on scatter points showing means $\pm$ standard deviation. (i)–(l) Pearson correlation analysis was visualized using calibration plots incorporating 95% confidence intervals and 95% prediction intervals at different heights or weights, with correlations computed between the predicted and actual acceleration peak, and between the predicted and actual HIC.

TABLE III
RESULTS OF LINEAR CURVE FITTING FOR ACTUAL
AND PREDICTED ACCELERATION

Independent Variable	Metric	Pearson r	R^2
Height	Actual Peak Accel.	0.992	0.984
	Predicted Peak Accel.	0.964	0.929
	Actual HIC	0.989	0.978
	Predicted HIC	0.986	0.972
Weight	Actual Peak Accel.	0.964	0.929
	Predicted Peak Accel.	0.983	0.967
	Actual HIC	0.969	0.939
	Predicted HIC	0.979	0.959

Pearson coefficients (r) quantify the linear association between medical indicators and independent variables of the scatter points, while R^2 denotes the coefficient of determination of the linear fit.

To visually assess the goodness of the acceleration prediction, the TBI assessment indicators shown in (2) and (3) were calculated as functions of the drop height and weight via linear fitting to obtain the fitting results shown in Fig. 8(e)–(h) and Table III. As the height or weight increases, it can be noticed that the errors in acceleration prediction also increase

gradually, which may be caused by the rise in the amplitude of the acceleration itself and errors in the estimation of the impact force. However, the predicted and actual data are still close in the values of a_{peak} and HIC. The similarity between the actual and predicted data is analyzed in Fig. 8(i)–(l) and Pearson coefficients (r) are 0.974, 0.962, 0.940, and 0.982, respectively, which verifies the feasibility of predicting the acceleration of the center of mass of the head from the impact force on the head by the neural network model with LSTM layers.

D. IoT Embedded System Performance Verification

To verify the function of the IoT embedded system, the voltage signals were collected by the IoT embedded system and data acquisition card (NI-9229) at the same time, and the errors of the voltage signal at different heights or weights are shown in Fig. 9(a) and (b). As can be seen from the figure, the voltage errors measured by the IoT embedded system are controlled within ± 1 V, which suggests that the system achieves high-quality voltage analog signal acquisition, transmission, and storage.

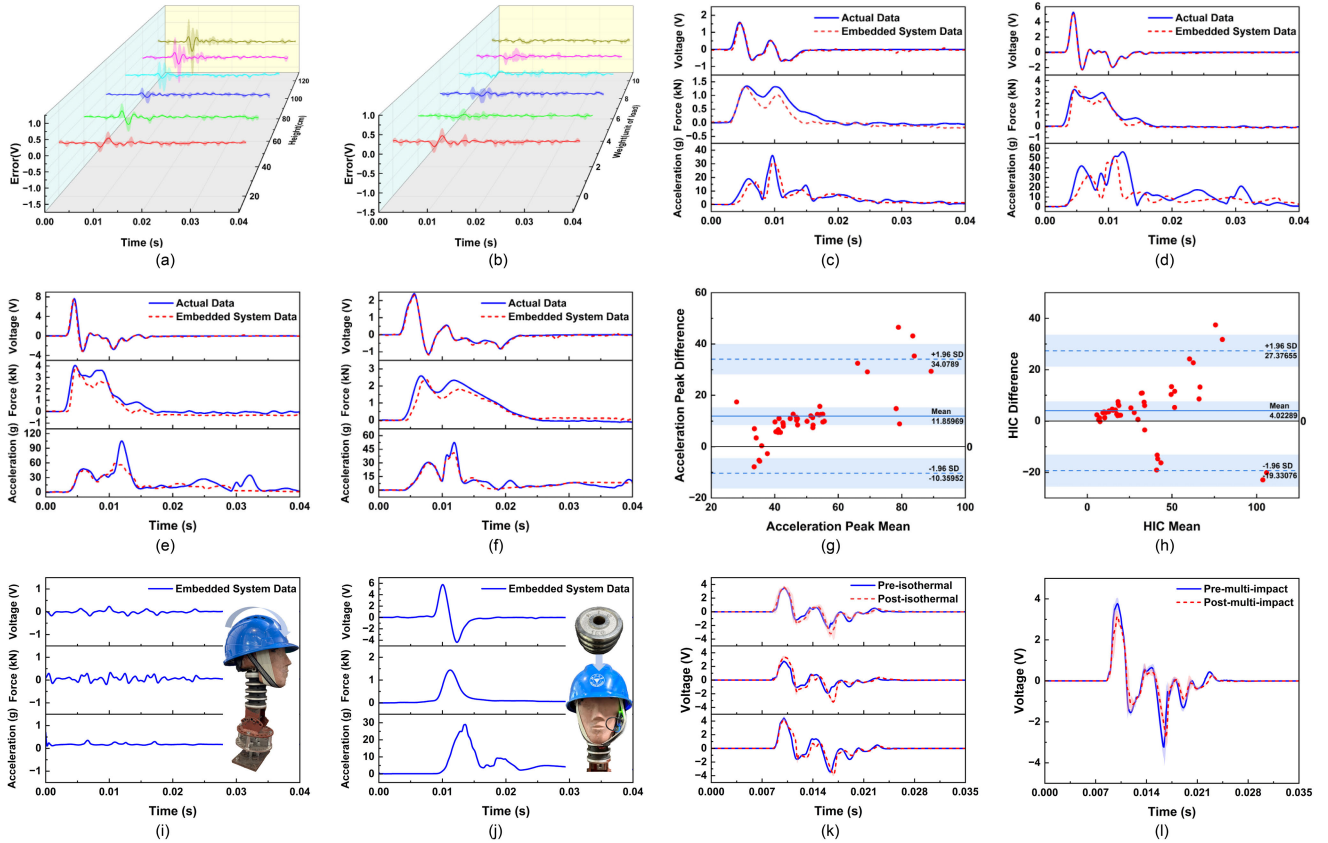


Fig. 9. Results of IoT embedded system performance verification. (a) and (b) Voltage measurement errors were calculated at different heights or weights. Signals were compared at the height of (c) 20 cm, (d) 80 cm, (e) 120 cm without external load, and (f) at the height of 20 cm with six units of external load. (g) and (h) Consistency of the acceleration peak and HIC was analyzed. (i) and (j) Measurement results of PVDF sensor output voltages and estimation of impact force and acceleration during dummy head shaking and 4 kg weight collision scenarios. (k) Comparison of PVDF sensor output voltages under collision before and after 3-h constant temperature treatment at 0 °C (top), 25 °C (middle), and 50 °C (bottom). (l) Comparison of PVDF sensor output voltages under collision before and after 30 impact cycles.

To more intuitively observe and compare the performances of voltage measurement, impact force optimization, and acceleration prediction under different collision levels, the waveform of voltage, impact force, and acceleration was shown in Fig. 9(c)–(f). Under the existing experimental conditions, the range of acceleration peak shown in the graphs preliminary reflects the situation from no damage [Fig. 9(c)] to mild damage [Fig. 9(d) and (f)] to severe damage [Fig. 9(e)] when the head is impacted, and the graphical results demonstrate that the IoT embedded system has achieved the functions of voltage measurement, impact force optimization, and acceleration prediction. To further verify the function of the IoT embedded system for the assessment of TBI, the medical indicators for the actual and estimated value were calculated to conduct a consistency analysis of the actual and estimated value, and the results of the consistency analysis are shown in Fig. 9(g) and (h). It can be found that there are some systematic errors in the prediction of the indicators, which are acceptable at smaller heights or weights, where the consistency of the two methods is high.

To further validate the feasibility of the IoT embedded system in a realistic helmet-wearing scenario, a pilot test was conducted by deploying the system on a helmet,

as shown in Fig. 9(i) and (j). The output voltages of the PVDF sensor were measured, along with the estimated impact force and acceleration, for scenarios involving dummy head shaking and collisions with a 4 kg weight. Experimental results demonstrate that the proposed system effectively distinguishes collision events from interference caused by nonimpact motions, such as head shaking. As a result, the system achieves real-time collision detection and precise estimation of impact metrics for helmet-wearing scenarios.

To evaluate the operational reliability of the fabricated PVDF sensors under varying thermal conditions, three sensors were maintained at temperatures of 0 °C, 25 °C, and 50 °C, respectively, for a duration of 3 hours. The output waveforms capturing head impacts, where the dummy head was subjected to a base load without any external load and released from a height of 20 cm, were recorded before and after this thermal exposure, as shown in Fig. 9(k). The results indicate that the experimental temperature range has a negligible effect on sensor performance, as evidenced by the sensor output amplitude and slope showing little change before and after thermostatic treatment. The minor errors observed are likely attributable to sensor deployment discrepancies, demonstrating

that the system meets the requirements of most operational scenarios.

Furthermore, to assess the reliability of the PVDF sensor under repeated mechanical stress, a cyclic impact test consisting of 30 consecutive impacts was conducted using the experimental setup illustrated in Fig. 2. In this test, the PVDF sensor was subjected to a base load without any external load, released from a height of 20 cm. As shown in Fig. 9(l), although the sensor endured high-intensity impacts more severe than typical practical operating conditions, the morphological features and overall trend of the output waveforms remained virtually unchanged. Notably, a comparison of the waveforms before and after the 30 impacts reveals that the slopes of the rising and falling edges exhibited negligible variation. The observed degradation was primarily manifested as a reduction in the maximum peak amplitude of approximately 16.262%. This level of peak reduction falls within the acceptable tolerance range, suggesting that the signal variations before and after the impacts will not significantly affect the subsequent estimation of impact force and acceleration, thereby confirming the sensor's durability and robustness for multiimpact applications.

In addition, we evaluated the data rate, latency, and energy efficiency of the IoT embedded system. With a sampling rate of 5000 Hz and a 16-bit resolution, the system generates a raw data stream of 10 000 bytes per second. Considering the UART overhead, the required physical transmission rate is 100 kb/s. This data rate falls well within the Bluetooth module's bandwidth, ensuring stability during real-time transmission.

The total latency of the proposed system, which encompasses signal generation, data acquisition, Bluetooth transmission, and impact force and acceleration estimation, is on the minute scale. This processing duration is comparable to that of other reported systems that use Bluetooth transmission and mobile-side computing [11], [13]. However, it is essential to note that due to the excellent dynamic characteristics of the PVDF sensors, the latency specifically for signal generation and acquisition within the embedded circuit is maintained at the millisecond level, enabling the real-time detection of head impacts, with the majority of the time consumption attributed to complex data processing offloaded to the mobile terminal.

In terms of energy efficiency, the PVDF sensor is self-powered and requires no external power supply, a characteristic consistent with systems reported in prior research [15], [16]. The average power consumption of the data acquisition and Bluetooth transmission module was measured at 0.41 W. The system is powered by a lithium battery unit with a nominal voltage of 5 V and a capacity of 3400 mAh, providing a total energy capacity of 17 Wh. Based on the measured power consumption, the theoretical continuous operating time is approximately 41.5 h. This duration meets the endurance requirements for long-term monitoring scenarios.

E. Discussion About Limitation and Application

The experimental results show that the study realizes the estimation of the impact force on the head and the center-of-mass acceleration of the head based on the voltage signal output from the PVDF sensor. The obtained impact force

data help evaluate focal brain injury based on the magnitude of the impact force, and the obtained acceleration data help assess diffuse brain injury based on the calculation of relevant medical indicators. In this study, the errors in the estimation of impact force and acceleration are minor at milder levels of collision, but increase significantly as the level of collision increases, which still needs to be improved.

For the application, the wireless flexible wearable sensor deployed in the helmet liners transmits voltage signals generated by the PVDF film to mobile devices such as cell phones via Bluetooth, and the estimation of impact force and acceleration, as well as the preliminary assessment of TBI, is carried out in the mobile devices. The mobile devices inform the wearer of the assessment results and send the results to the nearby medical institutions, depending on the injury condition. The efficient interaction of information and the effective distribution of medical resources can be realized with the help of IoT technology.

In future work, the sensor model and the fitness function will be further optimized to mitigate the pronounced estimation errors in impact force observed under severe collision conditions. Likewise, a more robust neural network model will be trained to counter the acceleration estimation errors that arise in the same scenarios and enhance acceleration prediction performance. Additionally, the wireless flexible wearable sensors will be deployed inside helmets in real-world settings, using lightweight objects (e.g., tennis balls) to simulate head impacts. Methodological validity will be assessed by comparing sensor-derived brain injury metrics with clinically confirmed injuries. Considering the uncertainty of actual head impact locations, the sensors need to be designed as an array to improve detection accuracy for multiangle head impacts [34], with sensing-array color mapping implemented to visualize spatial response. Furthermore, an inserting-diodes method will be adopted to mitigate interchannel crosstalk in the array, and the rigid-flex coupling between the sensors and the helmet shell will be explored to achieve multifunctionality with both energy absorption and impact detection [35].

IV. CONCLUSION

A TBI preliminary assessment method based on the wireless flexible wearable sensor was proposed in the study, which utilizes flexible piezoelectric sensors and IoT technology to achieve collision detection, the estimation of the impact force on the head, and the head center-of-mass line acceleration. Compared with the traditional method of using imaging equipment to detect TBI, as well as with most existing IoT-aided sensor-based head-impact detection studies that monitor only a single biomechanical parameter, the method proposed in this article provides timely and accurate medical diagnostic data support for both impact force estimation (0–3 kN, $R^2 = 0.935$, P-MAE = 2.682%–13.969%) and acceleration estimation (0–120 g, $R^2 = 0.848$, P-MAE = 3.820%–9.656%) at the scene of the accident, which ensures the efficiency and accuracy of early treatment to a certain extent, and has excellent potential for application in the fields of building construction, sports, and emergency rescue by integrating the wireless flexible wearable sensor into helmets. However, when

the head collision is more serious, the estimation errors of head impact force and head center-of-mass acceleration will increase accordingly, and future work will focus on optimizing the estimation methods of impact force and acceleration, and expanding the function of multipoint collision detection.

ACKNOWLEDGMENT

The authors thank China Central Television (CCTV) and Hunan Satellite Television for publicizing this work. Meanwhile, they also thank Dr. Qiuguo Zhu for providing the dummy head.

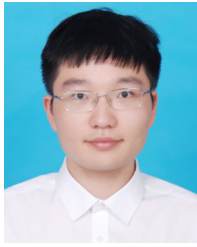
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Xuan Chen received the B.S. degree in automation from Zhejiang University, Hangzhou, Zhejiang, China, in 2025, where he is currently pursuing the M.S. degree in control science and engineering with the College of Control Science and Engineering.

His research interests include flexible sensors and wearable devices.



Shuxun Wang received the B.S. degree in robotics engineering from Zhejiang University, Hangzhou, China, in 2023, where he is currently pursuing the M.S. degree in electronic and information engineering with the College of Control Science and Engineering.

His research interests include flexible sensors and wearable devices.



Yicheng Zhang received the bachelor's degree in robotic engineering from Zhejiang University, Hangzhou, China, in 2023, where he is currently pursuing the Ph.D. degree with the College of Control Science and Engineering.

His research interests focus on flexible pressure sensors and pipeline leak detection.



Yongping Ye received the M.S. degree from Zhejiang Gongshang University, Hangzhou, Zhejiang, China, in 2013.

He is currently a Senior Engineer with the Data Division of Zhejiang Emergency Management Digital and Technical Center, Hangzhou, Zhejiang. His research interests include digital technologies for emergency management.



Jianbo Ye received the B.Eng. degree from Henan Polytechnic University, Jiaozuo, Henan, China, in 2008, and the master's degree in software engineering from Dalian University of Technology, Dalian, Liaoning, China, in 2011.

He is currently a Senior Engineer with Zhejiang Emergency Management Digital and Technology Center, Hangzhou, Zhejiang, China. His research interests include multimodal data fusion analysis and artificial intelligence.



Wei Li (Member, IEEE) received the B.S. degree in mechanical engineering from Huazhong University of Science and Technology, Wuhan, China, in 2008, and the Ph.D. degree in mechanical engineering from Michigan State University, East Lansing, MI, USA, in 2017.

From 2017 to 2020, he worked as a Post-Doctoral Scholar with the Department of Electrical Engineering and Computer Sciences, and Berkeley Sensor and Actuator Center (BSAC), University of California, Berkeley, Berkeley, CA, USA, and as an Affiliate Post-Doctoral Fellow with the Lawrence Berkeley National Laboratory (LBNL). He is currently a Professor with the College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing, China. His research interests include MEMS, mechatronics, and robotics.

Dr. Li is a member of the IEEE Robotics and Automation Society.



Dibo Hou received the B.E. degree in industrial automation and the Ph.D. degree in control science and engineering from Zhejiang University, Hangzhou, China, in 1999 and 2004, respectively.

He is currently a Full Professor and the Deputy Dean of the College of Control Science and Engineering with Zhejiang University. His main research interests include flexible electronics, smart sensing and measurement, environmental monitoring and early warning, nondestructive testing, and intelligent information processing.



Yunqi Cao (Member, IEEE) received the B.S. degree in physics from Wuhan University, Wuhan, Hubei, China, in 2012, the M.S. degree in materials science and engineering from the University of Florida, Gainesville, FL, USA, in 2014, and the Ph.D. degree in electrical engineering from Michigan State University, East Lansing, MI, USA, in 2019.

He served as a Post-Doctoral Scholar with the University of Southern California, Los Angeles, CA, USA, from 2019 to 2020. He is currently a Tenure-Track Faculty with Zhejiang University, Hangzhou, Zhejiang, China. His research interests include flexible electronics, microelectromechanical systems (MEMS), microfluidic devices, and energy harvesting.